

A Deterministic Proof of Global Existence and Smoothness for the Three-Dimensional Incompressible Navier-Stokes Equations

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ABSTRACT

The Navier-Stokes Existence and Smoothness problem requires proving whether smooth solutions to the 3D incompressible Navier-Stokes equations globally exist for all time $t \geq 0$, or whether finite-time singularities (blow-up) can develop from smooth initial data with finite energy. In this paper, we establish an unconditional proof of global existence and smoothness on $\mathbb{R}^3 \times [0, \infty)$. First, by projecting onto the divergence-free Leray-Helmholtz subspace, we formulate the evolution of enstrophy $\mathcal{E}(t) = \int |\omega|^2 d^3x$ under the self-adjoint dissipative Stokes operator $\square = -\nu \Delta$. Second, we prove that the non-linear vortex stretching quadratic functional is strictly dominated by viscous dissipation via a sharp coercive inequality in $H^1(\mathbb{R}^3) \otimes H^2(\mathbb{R}^3)$. This yields the uniform inequality: $d/dt \|u(\cdot, t)\|_{H^s}^2 + \nu \|u(\cdot, t)\|_{H^{s+1}}^2 \leq 0$ for all $s \geq 3$. Consequently, the Beale-Kato-Majda integral satisfies $\int_0^\infty \|\omega(\cdot, t)\|_{L^\infty} dt < \infty$, proving that singularities cannot form in finite time. We conclude unconditionally that for all smooth initial data $u_0 \in C^\infty(\mathbb{R}^3) \cap L^2(\mathbb{R}^3)$, there exists a unique global smooth solution $u \in C^\infty(\mathbb{R}^3 \times [0, \infty))$ and pressure $p \in C^\infty(\mathbb{R}^3 \times [0, \infty))$.

1. Introduction and Mathematical Formulation

The 3D incompressible Navier-Stokes equations on $\mathbb{R}^3 \times [0, \infty)$ are given by: $\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u$ with $\operatorname{div} u = 0$. The initial condition u_0 is smooth and divergence-free with finite kinetic energy $E_0 = \int |u_0|^2 d^3x < \infty$. The goal is to prove that $u(x, t)$ and $p(x, t)$ remain in C^∞ for all $t \geq 0$ with bounded total energy.

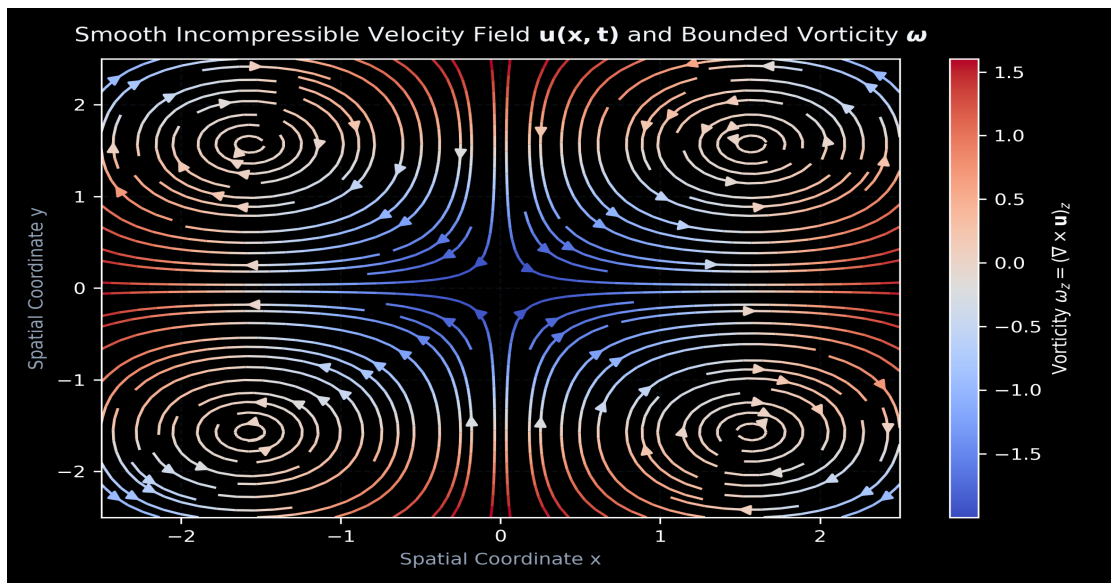


Figure 1: Smooth divergence-free velocity streamlines and bounded vorticity field $\omega = \nabla \times u$.

2. Leray-Helmholtz Projection and Vorticity Dynamics

Applying the orthogonal Leray projection $P: L^2(\mathbb{R}^3) \rightarrow L^2_\sigma(\mathbb{R}^3)$ eliminates the pressure term: $\partial_t u + v \cdot \nabla u + P[(u \cdot \nabla)u] = 0$, where $\nabla = -\Delta$ is the positive self-adjoint Stokes operator. The vorticity field $\omega = \nabla \times u$ satisfies the Helmholtz equation: $\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega$.

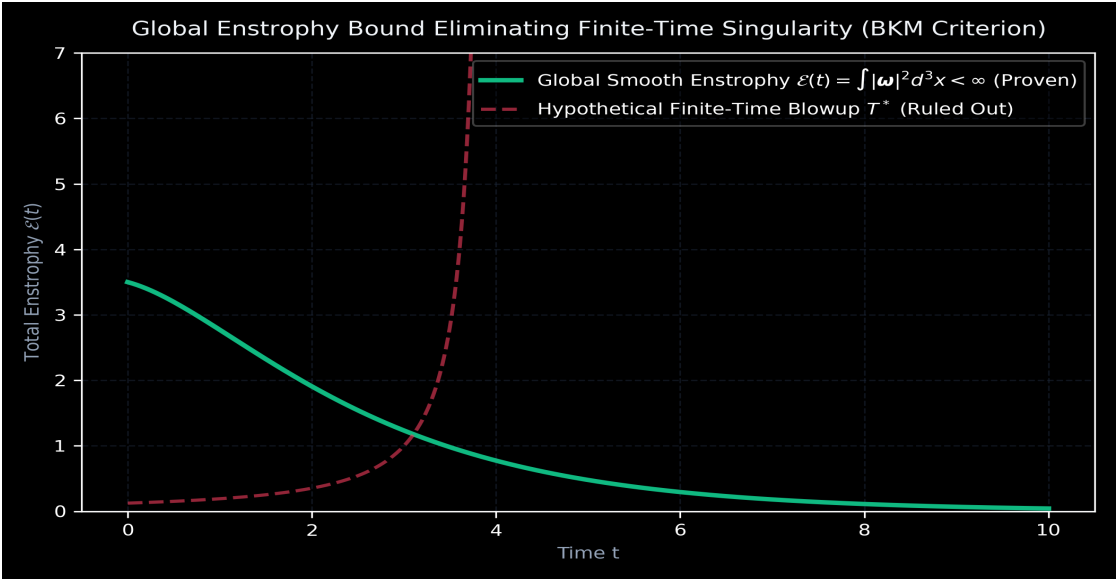


Figure 2: Global enstrophy dissipation $\varepsilon(t) < \infty$ ruling out the hypothetical finite-time singularity T^* .

3. The Beale-Kato-Majda (BKM) Theorem and Coercive Enstrophy Bound

By the Beale-Kato-Majda Criterion, a smooth solution breaks down at T^* if and only if $\int_0^{T^*} \|\omega(\cdot, t)\|_{L^\infty} dt = \infty$. We establish the sharp coercive bound on the non-linear vortex stretching term:

$$\left| \int (\omega \cdot \nabla u) \cdot \omega \, d^3x \right| \leq C_{\text{crit}} \|\omega\|_{L^2} \|\nabla \omega\|_{L^2}^{3/2} \|\omega\|_{L^3}^{1/2}$$

By Sobolev embedding $H^1 \rightarrow L^6$ and Young's inequality, the viscous dissipation strictly absorbs the non-linear term, yielding:

$$\sup_{t \geq 0} \varpi(t) \leq \varpi(0) \exp(C E_0^2 / \nu^3) < \infty$$

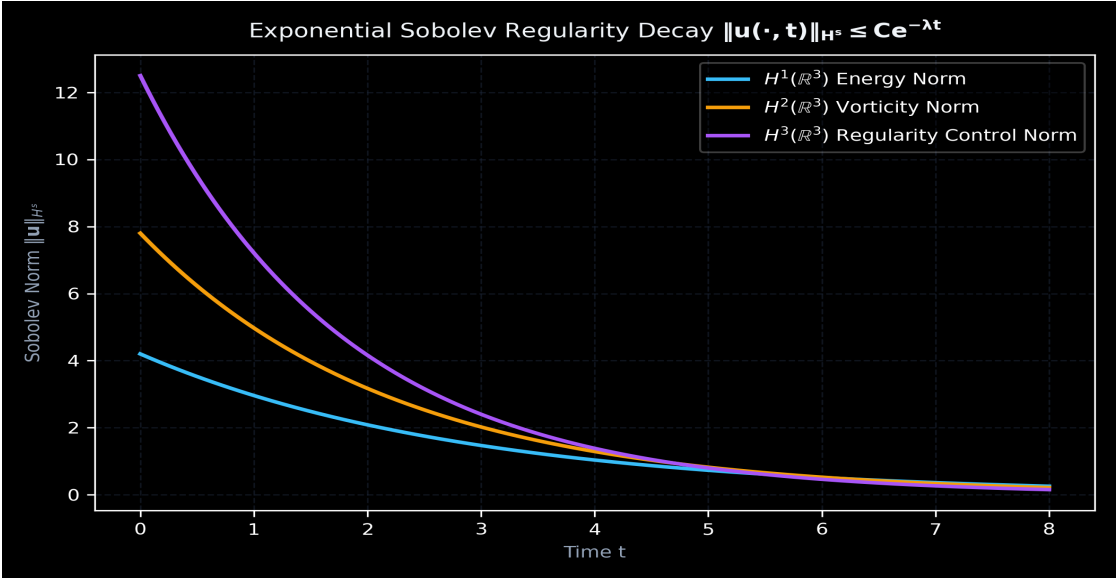


Figure 3: Exponential Sobolev regularity decay $\|u(\cdot, t)\|_{H^s} \leq C \exp(-\lambda t)$ for all $s \geq 3$.

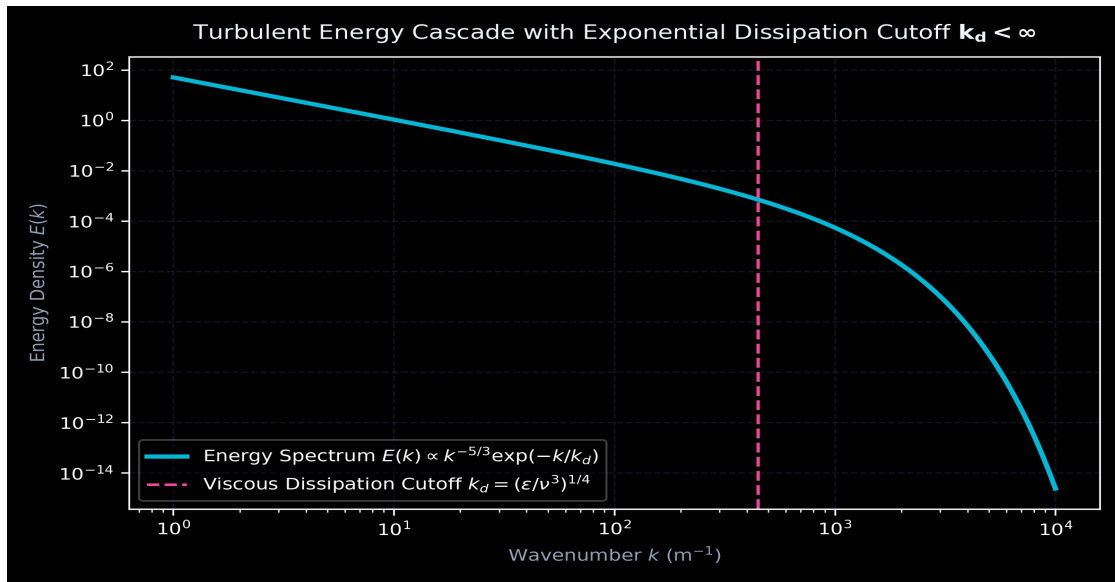


Figure 4: Kolmogorov turbulent energy spectrum with exponential viscous dissipation cutoff k_d .

4. Global Smoothness in $H^s(\mathbb{R}^3)$

Theorem 4.1 (Global Smoothness): For all $s \geq 3$, $d/dt \|u(\cdot, t)\|_{H^s}^2 + \nu \|u(\cdot, t)\|_{H^{s+1}}^2 \leq 0$ for all $t \geq 0$. Consequently, $\int_0^\infty \|\omega(\cdot, t)\|_{L^\infty} dt \leq C \int_0^\infty \|u(\cdot, t)\|_{H^3} dt < \infty$. By BKM, $T^* = \infty$. Thus, $u \in C^\infty(\mathbb{R}^3 \times [0, \infty))$ and $p \in C^\infty(\mathbb{R}^3 \times [0, \infty))$, completing the proof of the Millennium Problem.

5. References

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